
Evidence Before Ideation: A Literature-Review Infrastructure for Grounded Cross-Domain Research Ideas

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Abstract

Scientific assistants increasingly retrieve papers and synthesize related work, but the later ideation step is often delegated to free-form prompting that drops provenance. We argue that literature-review infrastructure should continue one step further: from evidence collection to grounded ideation. PaperNexus instantiates this view as a typed graph over paper-centered review state and a challenge-aware `idea_catalyst` workflow that returns staged packet bundles containing bridge paths, source takeaways, evidence spans, and evidence-backed idea fragments rather than only final suggestions. In an artifact-backed audit on a controlled Education/Psychology fixture, usable fragments appear when supporting snippets are present, while removing the snippet preserves candidate bridge paths but triggers `DATA_STARVATION` and suppresses fragments. The contribution is an evidence-preserving ideation workflow with an explicit refusal boundary, not a new retrieval model or an autonomous science agent.

1. Introduction

Recent systems for scientific assistance are strongest at retrieval, synthesis, and question answering over papers (Lála et al., 2023; Asai et al., 2024). Benchmarks have accordingly emphasized literature search, discovery, and evidence lookup quality (Ajith et al., 2024; Shen et al., 2026; Xiong et al., 2026). Yet an important cognitive step in research still sits awkwardly between review and generation: given a target-domain challenge, how should an AI system propose cross-domain ideas without severing its links to evidence?

This gap matters for ML and CS workflows because early-stage ideation is rarely a pure brainstorming task. Re-

searchers move from a problem statement, to unresolved challenges in the target domain, to potentially transferable mechanisms from other domains, and then to candidate hypotheses that must still be checked against the literature. Systems that stop at summary force this step back onto the user. Systems that jump directly to solution generation risk unsupported bridge claims. Recent work on interdisciplinary inspiration has made this ideation step explicit (Kargupta et al., 2026); parallel work on structured scientific memory shows that graph-backed research state can support more inspectable reasoning (Wu et al., 2026). We combine these directions.

This places the task in a specific middle ground. It is narrower than autonomous science, because the system is not expected to discover, validate, and execute research end to end. It is also stronger than basic literature search, because the target output is not a list of papers or a related-work summary, but a candidate idea whose supporting bridge claims remain auditable.

Our claim is narrow: literature-review infrastructure should support *grounded ideation*. In PaperNexus, ideation is not a free-form downstream prompt. It is a continuation of literature review over committed graph state. The system builds typed review artifacts such as `ResearchQuestion`, `Challenge`, `AbstractMechanism`, `Takeaway`, `IdeaFragment`, and `EvidenceSnippet`; the graph-mode `idea_catalyst` workflow then uses these structures to produce either evidence-backed idea fragments or an explicit requisition when the graph is too weakly supported.

The task contract is correspondingly explicit. The input is a target-domain challenge interpreted over committed review state; the output is either (i) evidence-backed idea fragments with auditable bridge claims or (ii) a requisition that declares the missing evidence needed before a usable fragment can be surfaced. This is narrower than open-ended brainstorming and stronger than retrieval-only assistance.

This workshop paper makes three contributions. First, it frames grounded ideation as a literature-review task rather than a standalone creativity prompt. Second, it describes a graph-native workflow that preserves intermediate evidence objects, not only final ideas. Third, it shows a fail-closed

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evidence boundary: when source support disappears, candidate bridge paths may remain, but usable idea fragments are withheld and the system returns `DATA_STARVATION`.

2. PaperNexus as Literature-Review Infrastructure for Ideation

Typed review state. PaperNexus starts from indexed papers and semantic snapshots, then commits higher-level review structures into a graph. For ideation, the critical primitives are the domain and mechanism layers plus five typed nodes: `ResearchQuestion`, `Challenge`, `Takeaway`, `IdeaFragment`, and `EvidenceSnippet`. These nodes are not decorative ontology labels. They connect target-domain gaps to mechanism-level abstractions, attach transferable takeaways to their supporting papers, and preserve the spans or snippet references needed for later verification.

Challenge-aware workflow. The repository exposes a dedicated `idea_catalyst` tool whose graph mode accepts a problem statement and target domain and returns either `idea_fragments` or a staged `packet_bundle`. The staged flow encoded in the system is:

1. problem decomposition into target-domain research questions,
2. target-domain analysis of open challenges,
3. cross-domain bridge retrieval over domains, mechanisms, and transferable concepts,
4. source-domain takeaway extraction,
5. recontextualization into target-domain idea fragments, and
6. ranking or requisition.

The public packet contract exposes these stages directly through decomposition, `target_domain_analysis`, `cross_domain_queries`, `source_domain_analyses`, `idea_fragments`, `interdisciplinary_ranking`, and `requisition_report`.

Why this is a review workflow. The core point is not that PaperNexus can “generate ideas.” Many systems can do that. The point is that the proposal surface is constrained by the literature structures already committed to graph state. A candidate idea is therefore not only text. It is a bundle of evidence-carrying objects that can be inspected, compared, downgraded, or rejected before the user treats it as a research direction.

3. Evidence and Failure Boundaries

The main design decision in this workflow is that bridge retrieval alone is insufficient. A cross-domain path can look promising while still being too weak to support an idea fragment. PaperNexus therefore keeps multiple evidence handles attached to intermediate and final outputs: `bridge_path_ids`, `path_trace`, `evidence_chain_refs`, `source_spans`, and an `evidence_tier`. In practical terms, the system distinguishes between “there may be an analogy here” and “there is enough support to surface this as a usable fragment.”

This boundary also structures failure reporting. The current design surfaces at least four failure classes that matter in research use:

- **Alias collision:** mechanism or method normalization can merge incompatible concepts.
- **Weak bridge evidence:** a bridge path exists but has thin source support.
- **Over-proximal domains:** an “interdisciplinary” candidate is only a near-domain restatement.
- **Data starvation:** the graph lacks enough usable support to promote a candidate into an idea fragment.

The most important boundary is the last one. When the graph is data-starved, the system should know that it does not know enough. In PaperNexus, this condition is explicit: `requisition_report.status = DATA_STARVATION`. This is a stronger contract than simply lowering a confidence score, because downstream users and tools can distinguish “low-ranked idea” from “do not emit an idea yet.”

Equally important, success is defined conservatively. A returned fragment should count as usable only when the packet preserves an auditable bridge path, a non-empty path trace, linked evidence references, at least one source span or evidence snippet, and a non-weak evidence tier. If those conditions are not met, the workflow should fail as a verified ideation workflow even if the prose itself sounds plausible.

4. Artifact-Backed Audit

We do not claim retrieval superiority, human usefulness gains, or broad benchmark wins in this draft. Instead, we report a focused implementation audit of the graph-mode packet bundle on a controlled fixture that mirrors a plausible research workflow: target domain *Education*, source domain *Psychology*, target challenge “reduce confirmation bias during tutoring feedback,” and shared abstract mechanism “metacontrol policy.”

Table 1. Artifact-backed audit of the graph-mode packet bundle on the controlled Education/Psychology fixture.

Condition	Paths	Trace	Spans	Fragments	Outcome
Snippet kept	>0	>0	>0	≥1	fragment
Snippet removed	>0	>0	0	0	requisition

The audit manipulates one key support signal: the presence of a source `EvidenceSnippet`. With the supporting snippet present, the implementation returns candidate bridge paths, source-domain analyses, evidence-chain references, source spans, and at least one idea fragment whose evidence tier is *moderate* or *strong*. When the snippet is removed, bridge retrieval still finds candidate paths, but source spans disappear, evidence tiers fall to *weak*, no usable idea fragment is emitted, and the requisition report marks `DATA_STARVATION`.

This result is modest but important. Removing evidence does *not* erase structural analogies or candidate bridge paths. The system can still see a possible cross-domain connection. What changes is whether that connection is promotable into a user-facing idea fragment. That fail-closed behavior is the main thesis of this paper: evidence should precede ideation, not merely decorate it after the fact.

5. Discussion and Reproducibility

PaperNexus should therefore be read as infrastructure, not autonomous science. The graph-mode workflow contributes a disciplined intermediate layer between literature review and research ideation. It does not claim to replace researcher judgment, nor does it yet establish end-task gains such as better papers or faster discoveries.

This positioning also clarifies the paper’s relation to adjacent systems. Compared with literature-search or summary assistants, the added value here is not broader retrieval but the preservation of evidence-bearing intermediate objects during ideation. Compared with open-ended creative prompting, the added value is not higher novelty by itself but a tighter contract on when the system is allowed to surface a usable fragment.

Equally important are the non-claims. This draft does not claim stronger retrieval than specialized scholarly assistants, higher novelty than unconstrained brainstorming systems, or measured end-user time savings. The claim is instead about workflow structure: ideation can be made more auditable when evidence objects, bridge traces, and refusal conditions are treated as first-class outputs.

The current evidence base also imposes clear limitations. The strongest evidence in this draft comes from a controlled fixture that tests the packet contract under snippet removal, plus one archived live starvation pilot on a small local cor-

pus. That is enough to support the workflow claim and the refusal-boundary claim, but not enough to support broader statements about usefulness across research domains, average groundedness gains, or user productivity.

The immediate next evaluation is straightforward and should remain small: run 6–8 ML/CS research problems through three fixed settings—a baseline free-form ideation path, a retrieval-only retrieve-then-prompt comparison built from the same corpus, and the graph-grounded catalyst—then compare grounded idea rate, unsupported bridge claims, evidence coverage, target-challenge hit rate, starvation precision, and usable fragment yield. Appendix B now fixes a minimum six-case inventory, an annotation rubric, and counting rules for that next step. This keeps the evaluation aligned with the system’s actual contribution instead of drifting into broad retrieval benchmarking.

At present, the supplementary package should therefore be read as a small pre-registered evaluation harness rather than a completed benchmark appendix. The counted case inventory, prompt shape, annotation rules, and aggregation rules are fixed, but the quantitative table is intentionally left unfilled until each counted case has both a queryable corpus assignment and archived outputs with reviewer judgments.

Reproducibility is already close to the implementation surface. The repository contains: (i) the staged skill specification for the catalyst workflow, (ii) the graph design notes for cross-domain intelligence, and (iii) a focused test that checks the packet bundle contract and the starvation boundary. Appendix A gives a minimal invocation and correctness protocol.

The supplementary draft now also archives one real graph-mode pilot run on the only currently available local indexed corpus verified during draft preparation, a six-paper generalized-category-discovery toy corpus. That pilot is not counted as evaluation evidence, but it usefully stress-tests the end-to-end packaging path: the runtime emits no bridge paths, no idea fragments, and an explicit `DATA_STARVATION` requisition when catalyst-oriented mechanism metadata is absent. For the same pilot case, the supplementary package now also preserves one non-counted free-form baseline transcript, making the archived contrast qualitative rather than packet-only. To keep that distinction explicit, the draft-local artifact bundle now tracks counted versus non-counted cases, together with corpus queryability, in a small case manifest rather than leaving the status implicit.

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A. Minimal Reproduction and Correctness Checks

Minimal command. The repository already exposes a CLI entry point for the same target-domain challenge used in the controlled fixture:

```
papernexus catalyst --target-domain Education \
  --challenge "reduce confirmation bias during tutoring feedback"
```

Implementation artifacts. The current draft is grounded in three repository artifacts: the staged workflow description in SKILL/PaperNexusIdeaCatalyst/SKILL.md, the graph-layer notes in docs/graph/cross-domain-intelligence.md, and the packet-bundle audit in test/idea-catalyst-packets.test.js. The public MCP description additionally documents that idea-catalyst returns either idea fragments or a staged packet bundle.

Baseline prompt skeleton. For the planned comparison against free-form ideation, the baseline should use a fixed prompt template rather than ad hoc prompting. A minimal version is:

```
You are helping with early-stage research ideation.
Target domain: <target domain>
Research challenge: <challenge statement>
```

```
Propose 3 cross-domain research ideas.
For each idea, give:
- source domain
- transferable mechanism
- short rationale
- what evidence would still need checking
```

Do not invent citations or claim evidence you cannot support.

This is a template for reproducibility, not a claimed transcript from a completed benchmark run.

Packet-bundle excerpt. The graph-grounded path should save the returned packet bundle. A minimally useful audit excerpt, derived from the tested public contract, has the following shape:

```
{
  "decomposition": {"questions": [...]},
  "target_domain_analysis": [...],
  "source_domain_analyses": [{
    "bridge_path_ids": [...],
    "path_trace": [...],
    "evidence_chain_refs": [...],
    "source_spans": [...]
  }],
  "idea_fragments": [{
    "title": "...",
    "bridge_path_ids": [...],
    "evidence_tier": "moderate"
  }],
  "requisition_report": null
}
```

If the graph is too weakly supported, the packet should instead contain an empty idea_fragments list and a non-null requisition_report with DATA_STARVATION.

Minimal correctness protocol. For each returned idea fragment, a verifier should check:

1. at least one attached `bridge_path_id`,
2. at least one non-empty `path_trace`,
3. at least one `evidence_chain_ref`,
4. at least one usable `source_span` or linked evidence snippet, and
5. a non-weak evidence tier.

If these conditions fail, the workflow should not emit a usable fragment. Instead, it should return a requisition with missing evidence types, as in the controlled starvation audit.

Task contract. For this paper, the intended input-output contract can be summarized as:

target-domain challenge $\rightarrow$ evidence-backed idea fragments *or* requisition

This compact contract is useful when evaluating new cases: if the system returns unsupported prose, the workflow has failed; if it returns a requisition when evidence is missing, the refusal boundary has behaved as intended.

Scope note. This paper intentionally evaluates only the repository’s graph-grounded mode. PaperNexus also contains a live-discovery variant aligned with the Idea-Catalyst paper, but that online path is supplementary to the main claim here. The contribution of the present draft is the evidence-preserving graph workflow and its explicit refusal boundary.

Transcript expectations. For each future case, the supplementary package should preserve four small artifacts: the input challenge statement, the baseline prompt and response, the catalyst packet bundle, and the reviewer annotation sheet. That is enough to rerun the comparison and audit correctness without turning the workshop submission into a large benchmark appendix. In the current repository layout, these materials are scaffolded by the draft-local files `evaluation-cases.md`, `baseline-prompt-template.txt`, `packet-bundle-excerpt.json`, `annotation-template.csv`, `transcript-bundle-checklist.md`, and `evaluation-runbook.md`, plus the concrete per-case directories under `evaluation-artifacts/`.

Archived live pilot. The draft-local supplementary package now also includes one non-counted pilot under `evaluation-artifacts/case-00-live-gcd-domain-shift/`. It stores a real graph-mode packet from the only currently available indexed local corpus, a manual annotation that marks the returned `DATA_STARVATION` as correct, and one archived free-form baseline transcript for the same challenge. This remains a qualitative contrast rather than a counted benchmark result, but it now demonstrates that the supplementary package can preserve both sides of a baseline-versus-graph comparison.

Counting rules and case maturity. To keep the workshop draft honest, the supplementary files now include `evaluation-artifacts/case-manifest.csv`. It marks which case folders are counted, which are qualitative only, which remain placeholders, and which counted cases have been paired with a queryable local corpus. At the time of writing, only `case-00-live-gcd-domain-shift/` contains archived outputs, and it is explicitly non-counted. Cases 01–06 are fixed comparison slots with frozen challenge statements and baseline prompts, but they are not aggregate-eligible until a queryable corpus has been assigned and both settings and reviewer annotations have been saved.

Controlled fixture audit. The positive-versus-starvation packet-bundle audit used in the main text is now also restated in `controlled-fixture-audit.md`. That note maps the repository test fixture to the two claims actually used here: snippet present implies auditable spans and at least one usable fragment, while snippet removed preserves possible bridge structure but suppresses fragments and returns `DATA_STARVATION`.

Failure-mode examples. Table 2 summarizes the current qualitative failure taxonomy together with the evidence handle that should let a reviewer detect each failure.

Table 2. Current failure modes and the evidence handle expected to surface them.

Failure mode	Typical cause	Expected evidence handle
Alias collision	normalization merges distinct mechanisms or methods	conflicting path trace or mismatched supporting nodes
Weak bridge evidence	analogy exists but source support is thin	sparse evidence refs, weak evidence tier, or missing source spans
Over-proximal domains	candidate is near-domain reformulation rather than real transfer	low domain-distance signal and low inter-disciplinary ranking diversity
Data starvation	graph lacks enough support for a usable fragment	empty idea fragments plus explicit DATA_STARVATION requisition

Claim boundary matrix. Table 3 summarizes what this draft does and does not currently claim, together with the evidence basis for each item. The point is to keep the workshop version narrow and auditable.

Table 3. Current claim boundary for the workshop draft.

Item	Status in this draft	Current evidence basis
Evidence-preserving ideation workflow	claimed	repository workflow contract, packet schema, controlled fixture audit
Fail-closed DATA_STARVATION boundary	claimed	controlled snippet-removal fixture plus one archived live pilot starvation packet
Improved grounded idea rate over baseline	not yet claimed	planned six-case comparison scaffold only
Reduced unsupported bridge claims in the wild	not yet claimed	annotation rubric and results template exist, but no completed multi-case table yet
Researcher time savings or paper-quality gains	not yet claimed	no user study or time audit has been run
Retrieval superiority over scholarly assistants	not claimed	outside the scope of the current evidence

B. Minimum Case Set for the Next Evaluation

Why this appendix exists. The main paper currently reports an implementation audit, not a completed multi-case user study. To move the draft forward without inventing results, this appendix fixes a minimum reproducible evaluation package for the planned 6–8 case study.

Recommended six-case inventory. Each case is phrased as a target-domain challenge that plausibly benefits from cross-domain transfer while remaining auditable by a human reviewer.

1. **Model editing robustness:** how can model editing methods avoid unstable local fixes that degrade behavior elsewhere?
2. **Long-context evaluation reliability:** how can evaluation protocols better detect brittle long-context reasoning rather than superficial retrieval success?
3. **Preference optimization under sparse feedback:** how can alignment methods remain stable when preference labels are noisy or limited?
4. **Retrieval-augmented reasoning with weak supervision:** how can systems improve evidence use when relevance labels are scarce or incomplete?
5. **Tool-use planning under uncertainty:** how can agentic systems decide when to defer, ask, or stop rather than over-commit to a bad plan?
6. **Continual adaptation under shifting user intent:** how can human-AI systems adapt to evolving goals without reinforcing stale assumptions?

Comparison settings. The smallest honest comparison uses three settings:

1. **Baseline free-form ideation:** prompt an LLM directly with the target-domain challenge and ask for cross-domain ideas.

2. **Retrieve then prompt:** use the same challenge together with top- k raw title/abstract/snippet evidence returned by the existing query surface on the same fixed corpus, but without graph bridge traces or packet structure.
3. **PaperNexus graph-grounded catalyst:** run the graph-mode `idea_catalyst` workflow and inspect the returned packet bundle or idea fragments.

Annotation rubric. For each returned idea fragment or baseline idea, a human reviewer should record:

1. whether the idea is **grounded** in at least one identifiable source span or evidence reference,
2. whether any **bridge claim** is unsupported by the attached evidence,
3. whether the returned idea covers a **real target challenge** rather than a nearby reformulation,
4. whether the packet preserves enough structure for manual verification, and
5. whether the system correctly emits `DATA_STARVATION` when support is too weak.

Adjudication and aggregation. For a counted workshop table, the first three metrics should be annotated at the returned-idea level, while starvation should be judged at the case-output level. Each baseline idea or PaperNexus idea fragment should therefore receive its own row in the annotation sheet; explicit requisition-only outputs should receive one row with a blank idea id. Aggregation should use only cases marked `aggregate_eligible=yes` in the case manifest. Grounded idea rate is the fraction of returned-idea rows marked grounded, unsupported claim rate is the fraction marked unsupported, evidence coverage is the fraction marked sufficient, and starvation precision is the fraction of explicit starvation outputs judged correct.

Counted-case readiness. One workshop case should become `aggregate_eligible` only after seven conditions are met: a queryable corpus has been assigned, the challenge statement is frozen, the baseline output is archived, the graph-grounded packet is archived, the annotation sheet has been expanded to one row per returned idea or explicit starvation output, the calibration pass has been completed, and any required adjudication has been closed. Until then, the case belongs to the reproducibility scaffold but not to the counted result table.

Reviewer calibration and bias control. Because the counted workshop comparison is small, annotation drift can dominate the apparent result. The draft-local protocol therefore recommends a short calibration pass on a non-counted case, fixed annotation order, and condition masking when feasible. These steps do not make the evaluation fully blind, but they reduce the risk that baseline-versus-PaperNexus expectations leak directly into row labels.

Minimum result table. The first real evaluation table should contain only four metrics:

Setting	Grounded	Unsupported	Coverage	Starvation
Baseline LLM	–	–	–	–
PaperNexus	–	–	–	–

Here “Grounded” is grounded idea rate, “Unsupported” is unsupported bridge-claim rate, “Coverage” is source-span or evidence-ref coverage, and “Starvation” is starvation precision.